

What Is the Best Location to Open a Car Repair Shop in Spain?

Caique Oliveira

2026-06-05

How to Read This Report

Sections 1 through 4 present the problem, the results, and the business recommendation. These sections are intended for readers primarily interested in the practical conclusions of the analysis.

The remaining sections describe the mathematical methodology, model derivation, and results analysis for readers interested in understanding the technical details of the study.

1 Context

The choice of location determines the success or failure of many small businesses. In the case of a car repair shop, the decision is especially complex because favorable business factors often point in opposite directions.

Large urban centers such as Madrid and Barcelona offer a vast consumer market. However, they impose some of the highest rents and cost of living in Spain. Inland cities such as Ciudad Real or Jaén have much lower operating costs, but serve a significantly smaller vehicle fleet.

Which of these situations is more advantageous? This work investigates precisely this balance, using public data and a mathematical model to generate an objective ranking of Spanish provinces with the greatest potential for establishing a car repair shop.

2 The Approach

The central question — which is the best province? — cannot be answered by observing a single variable. An index that combines multiple indicators coherently is necessary.

For this purpose, six data points were collected for each province: vehicle fleet, population density, per capita income, residential rent, consumer price index (CPI), and unemployment rate. From these indicators, a composite index S was constructed that follows that logic: the potential of a province is directly proportional to the size and wealth of its market, and inversely proportional to the costs that the business will face.

The index is given by the formula:

$$S = \frac{\sqrt{\text{Fleet}} \times \text{Income} \times \text{Density} \times (1 - \text{Unemployment})}{\text{Rent} \times \text{CPI}} \quad (1)$$

This structure captures important economic relationships. The use of the square root on the fleet reflects the reality that the marginal return of an additional vehicle is decreasing: having many cars is an advantage, but the gain is smaller as the market is already large. Income and density enter directly, as they are associated with higher demand for services. Unemployment appears as an attenuation factor, penalizing provinces where the population's purchasing power is structurally lower.

In the denominator, rent and cost of living represent the main forces that drain the shop's profit. The more expensive they are, the lower the net potential of the business. The result is an index that reflects profit potential realistically, without arbitrary weights.

3 The Solution

The index was calculated for the 50 continental provinces of Spain, excluding Ceuta and Melilla because they are *outliers* with exceptional densities and unemployment. The results were normalized by the maximum value (Madrid = 1.000) to facilitate comparison.

Figure 1 shows the result for the top 15 ranked provinces. Madrid (1.000) and Barcelona (0.651) lead the ranking, as expected, due to the combination of an immense fleet, high density, and high income. The most valuable result, however, lies in the following positions.

Provinces such as Bizkaia (Basque Country), Valencia, Gipuzkoa, and Pontevedra emerge as robust alternatives. They offer significant market potential (S values between 0.15 and 0.35) while presenting considerably lower cost structures than Madrid. In these provinces, the balance between potential revenue and operating cost is particularly favorable.

It is notable that some high-scoring provinces, such as Alicante, Malaga, the Balearic Islands, and the Canary Islands, have economies strongly dependent on tourism. This is a risk factor that the quantitative model does not capture,

but that a prudent entrepreneur should consider, given the seasonality and vulnerability of the tourism sector to external crises.

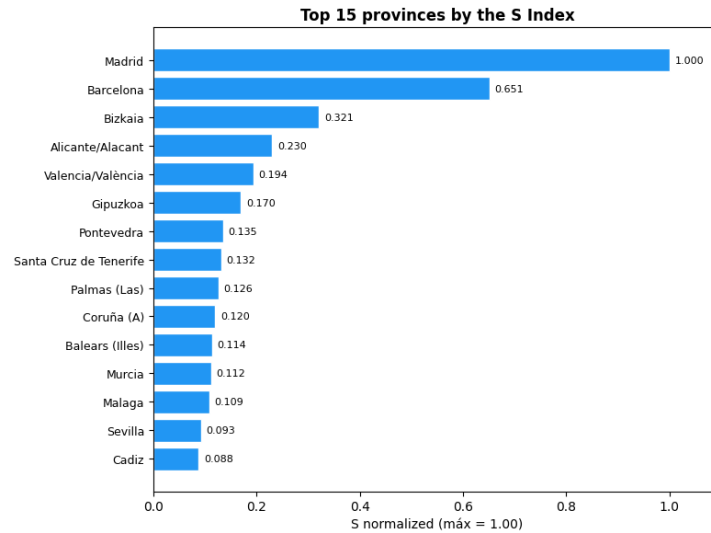


Figure 1: The 15 most promising provinces for opening a car repair shop.

4 Strategic Recommendation

Based on the index S and the qualitative risk analysis, the recommendation for the entrepreneur is to focus on a specific set of provinces.

Main Recommendation: The provinces of **Bizkaia**, **Valencia/València**, **Gipuzkoa**, and **Pontevedra** represent the best compromise between profit potential and business stability.

In these regions, the economy is diversified, not dependent on a single volatile sector such as tourism. The fleet is large enough to ensure stable demand, while operating costs are significantly lower than in Madrid or Barcelona. The recommendation is that the final prospecting be done *in loco* in the capitals of these provinces and in nearby medium-sized cities, seeking available warehouses and observing the presence of direct competitors.

Note: The analysis above provides all the information needed for decision-making. The following sections cover the technical methodology for transparency and professional documentation purposes.

5 Modeling Process

The starting point is the definition of the index S as a balance between market forces (numerator) and cost forces (denominator). Each variable was collected from official sources, as detailed in Table 1.

Table 1: Variables used in the model.

Variable	Source
Vehicle fleet (passenger cars)	Dirección General de Tráfico (DGT)
Population density (inhabitants/km ²)	Instituto Nacional de Estadística (INE)
Average net income per person (€/year)	INE (Atlas de Distribución de Renta)
Average residential rent price (€/m ² /month)	Idealista and Fotocasa portals
Consumer Price Index (CPI)	INE
Unemployment rate (%)	INE (Encuesta de Población Activa)

To ensure that the vehicle fleet did not dominate the index disproportionately, its square root was used. This reflects the hypothesis that the marginal value of an additional vehicle for a shop with fixed capacity is decreasing, modeling the real customer potential more faithfully.

The final index for each province i is calculated as:

$$S_i = \frac{\sqrt{\text{Fleet}_i} \times \text{Income}_i \times \text{Density}_i \times (1 - U_i)}{\text{Rent}_i \times \text{CPI}_i}$$

where U_i is the unemployment rate expressed as a decimal (e.g., 5% = 0.05). The final scores are normalized by dividing all values by the largest S found, so that the leader receives 1.00 and the others receive a value between 0 and 1.

6 Implementation

The index calculation was implemented in Python with the `numpy` library. The data collection process was the longest and most complex step, requiring manual compilation of information from multiple official sources.

```
import numpy as np

# 50 continental provinces (excluding Ceuta and Melilla)
provinces = [
    "Araba/Alava", "Albacete", "Alicante/Alacant", "Almeria", "
    Avila", "Badajoz",
    "Balears (Illes)", "Barcelona", "Burgos", "Caceres", "Cadiz
    ", "Castellon/Castello",
    "Ciudad Real", "Cordoba", "Coruna (A)", "Cuenca", "Girona",
    "Granada",
```

```

"Guadalajara", "Gipuzkoa", "Huelva", "Huesca", "Jaen", "
Leon", "Lleida",
"Rioja (La)", "Lugo", "Madrid", "Malaga", "Murcia", "
Navarra", "Ourense",
"Asturias", "Palencia", "Palmas (Las)", "Pontevedra", "
Salamanca",
"Santa Cruz de Tenerife", "Cantabria", "Segovia", "Sevilla"
, "Soria",
"Tarragona", "Teruel", "Toledo", "Valencia/Valencia", "
Valladolid", "Bizkaia",
"Zamora", "Zaragoza"
]

# Vehicle fleet (passenger cars)
fleet = np.array([
170126, 214826, 1129748, 422961, 99630, 418171, 768520,
2452618, 209409, 242376,
631949, 340459, 277535, 417331, 667450, 126047, 447302,
518212, 170792, 344787,
288360, 134476, 329191, 283136, 251717, 165966, 220871,
4171515, 957415, 865790,
362771, 198628, 550797, 98180, 685799, 572787, 188894,
636391, 336037, 101208,
1022948, 55531, 457395, 84631, 465698, 1352901, 283734,
541360, 109817, 456019
])

# Average net income per person (EUR/year)
income = np.array([
17806, 13058, 12313, 11543, 13602, 12068, 16118, 17062,
15988, 12771,
12335, 14029, 12529, 12386, 15319, 12725, 15130, 12709,
14689, 19616,
12008, 15140, 11847, 15110, 14713, 14827, 14316, 18142,
12950, 12446,
16423, 14049, 15784, 15147, 13806, 14137, 14583, 13366,
15043, 14949,
12964, 15626, 14559, 14573, 12422, 14435, 15590, 18738,
13897, 15650
])

# Population density (inhabitants/km^2)
density = np.array([
112.6, 26.2, 349.5, 87.8, 20.0, 30.6, 250.4, 770.6, 25.4,
19.5,
169.5, 94.6, 25.0, 56.1, 142.8, 11.7, 140.6, 74.8, 23.4,
370.1,
53.2, 14.7, 45.8, 28.8, 37.7, 64.8, 33.1, 886.2, 245.1,
140.2,
65.8, 42.0, 95.7, 19.7, 287.9, 210.8, 26.6, 322.1, 111.4,

```

```

        22.9,
        140.9, 8.7, 138.8, 9.2, 49.1, 255.6, 65.2, 526.8, 15.7,
        57.8
    ])

    # Average residential rent price (EUR/m^2/month)
    rent = np.array([
        12.1, 8.0, 12.0, 8.5, 7.8, 7.1, 19.1, 20.4, 9.4, 7.4,
        10.4, 8.8, 7.1, 8.6, 9.5, 7.5, 12.6, 9.9, 10.0, 16.4,
        9.0, 10.1, 6.3, 7.8, 11.5, 9.1, 7.5, 20.8, 16.6, 9.0,
        11.1, 7.5, 10.2, 7.8, 15.4, 10.4, 9.5, 15.1, 10.9, 12.4,
        11.7, 8.5, 10.1, 7.4, 8.7, 13.6, 9.2, 14.7, 7.1, 10.7
    ])

    # Consumer Price Index
    cpi = np.array([
        120.220, 121.476, 120.291, 120.524, 120.713, 120.159,
        120.332, 118.829, 120.713, 120.159,
        120.524, 120.291, 121.476, 120.524, 120.469, 121.476,
        118.829, 120.524, 121.476, 120.220,
        120.524, 120.016, 120.524, 120.713, 118.829, 120.117,
        120.469, 119.383, 120.524, 119.672,
        120.082, 120.469, 119.791, 120.713, 119.798, 120.469,
        120.713, 119.798, 119.991, 120.713,
        120.524, 120.713, 118.829, 120.016, 121.476, 120.291,
        120.713, 120.220, 120.713, 120.016
    ])

    # Unemployment rate (%)
    unemployment = np.array([
        5.03, 11.96, 13.28, 14.46, 11.30, 17.59, 11.38, 8.44, 8.41,
        13.96,
        20.82, 13.97, 16.27, 15.43, 8.44, 11.30, 9.28, 21.34, 9.31,
        6.35,
        20.57, 10.85, 18.95, 10.34, 5.42, 9.52, 7.35, 9.72, 16.49,
        11.32,
        9.32, 9.77, 11.81, 6.92, 15.48, 10.41, 10.56, 16.94, 7.48,
        8.01,
        15.69, 5.36, 14.45, 9.46, 11.84, 11.93, 8.62, 6.69, 15.41,
        6.96
    ])

    # Convert unemployment to decimal rate
    U = unemployment / 100.0

    # Calculate the S index
    #  $S = \sqrt{\text{Fleet}} * \text{Income} * \text{Density} * (1 - U) / (\text{Rent} * \text{CPI})$ 
    numerator = np.sqrt(fleet) * income * density * (1 - U)
    denominator = rent * cpi
    S = numerator / denominator

```

```

# Normalize by maximum (Madrid = 1.000)
S_norm = S / np.max(S)

# Obtain ranking (descending order)
ranking = np.argsort(S_norm)[::-1]

# Print top 15
for i, idx in enumerate(ranking[:15], 1):
    print(f"{i:2d}. {provinces[idx]:25s} S_norm={S_norm[idx]:.3f}")

```

Listing 1: Calculation of the S index for the 50 continental provinces

7 Results

Table 2 presents the 15 provinces with the highest scores on the normalized index S .

Table 2: Top 15 provinces by normalized index S .

Position	Province	S_{norm}
1	Madrid	1.000
2	Barcelona	0.651
3	Bizkaia	0.321
4	Alicante/Alacant	0.230
5	Valencia/València	0.194
6	Gipuzkoa	0.170
7	Pontevedra	0.135
8	Santa Cruz de Tenerife	0.132
9	Palmas (Las)	0.126
10	Coruña (A)	0.120
11	Balears (Illes)	0.114
12	Murcia	0.112
13	Malaga	0.109
14	Sevilla	0.093
15	Cadiz	0.088

8 Limitations and Future Work

The analysis has some important limitations that the decision-maker should be aware of:

- **Local competition:** the absence of public data on the exact location of competing shops prevented market saturation from being included in

the model. The analysis considers only potential demand, not the already installed supply.

- **Single-theme economy:** provinces strongly dependent on tourism (such as Alicante, the Balearic Islands, and the Canaries) obtained good scores, but present a risk of seasonality and instability that the index does not capture.
- **Micro-location:** the final choice within a province depends on factors such as warehouse availability, the exact price of the location, and local traffic characteristics. This decision needs to be made *in loco* by the entrepreneur.

Future work could incorporate competition data via geolocation, estimates of vehicle wear based on climate and topography, and a more detailed multicriteria analysis for the final choice of municipality. The presented model constitutes a solid and objective basis for the first stage of the decision.

9 Conclusion

The index S synthesized six socioeconomic variables into a single indicator of potential for opening a car repair shop. Applied to the provinces of Spain, it revealed that, although Madrid and Barcelona are the largest markets, the provinces of **Bizkaia, Valencia/València, Gipuzkoa, and Pontevedra** offer the most advantageous balance between market size and operating costs, in addition to having more diversified and stable economies.

The complete process demonstrated that the success of a data analysis project depends as much on mathematical sophistication as on the honesty in recognizing the limits of available information. The final result is a clear and well-founded recommendation, ready to guide the entrepreneur's decision.